Venkat Sai Subash Panchakarla

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Education

University of Cincinnati April 2026
Master of Science in Information Technology (GPA: 4.00 / 4.00) *Cincinnati, Ohio*

- **Relevant Coursework:** Machine Learning & Data Mining, Deep Learning, Natural Language Processing, Cloud Computing, Azure Data Engineering

Technical Skills

Languages: Python, SQL, TypeScript, R, JavaScript
Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Neural Networks, NLP, Transformers, Feature Engineering, Model Evaluation
Generative AI: LLMs, RAG, LangChain, LangGraph, Hugging Face, Prompt Engineering, Embeddings, Vector Databases, Agentic AI
MLOps & Cloud: AWS, SageMaker, Bedrock, Docker, Kubernetes, MLflow, CI/CD, Model Deployment, Model Monitoring
Frameworks & Tools: FastAPI, Flask, REST APIs, Git, GitHub Actions, Elasticsearch, PostgreSQL, FAISS

Experience

University of Cincinnati Mar 2025 - Sep 2026
Data Science Engineer *Cincinnati, Ohio*

- Designed and implemented an unsupervised deep learning Autoencoder model to analyze crime similarity probabilities, leveraging reconstruction error to identify patterns across a 2M+ record dataset.
- Built a robust data preprocessing and feature engineering pipeline, including missing-value imputation, one-hot encoding, and normalization of demographic and geographic variables to ensure high-quality model inputs.
- Developed a scalable real-time scoring framework to evaluate new incidents, enabling probabilistic crime assessment and supporting data-driven public safety and research initiatives.

Cybereconn Apr 2022 – Nov 2024
Junior AI/ML Engineer *Gurugram, Haryana*

- Developed an AI-based coding assistance tool that generates multiple high-quality examples per topic, helping learners better understand programming concepts through structured demonstrations.
- Applied machine learning and NLP workflows across projects, including prompt design, document ingestion, semantic search, and response generation, to deliver scalable and user-centric AI solutions.
- Designed and developed Vartalap India, a multi-use AI-powered content generation platform, enabling automated creation of Instagram captions, product descriptions, emails, blogs, translations, and content summaries using NLP techniques.

Projects

LLM-Based Research Paper Analysis Engine | *Python, LangChain, FastAPI, NLP*

- Developed AI platform for summarizing, clustering, and comparing research papers using LLM-driven semantic analysis with LangChain and FastAPI, cutting manual review effort by 75%.

Crime Similarity & Risk Prediction System | *AutoEncoders, Scikit-Learn, Python, Deep Learning*

- Built unsupervised deep learning system using AutoEncoders across 2M+ records, improving pattern detection accuracy by approximately 35% and enabling real-time risk scoring.

CareerPilot AI | *Python, LangChain, Llama3.1, Streamlit*

- Built AI-powered job application assistant using LangChain, Ollama Llama 3.1, Streamlit, and SQLite; integrated LLM-driven visa sponsorship classification, H1B lookup, and ATS score calculation, reducing manual job research effort by approximately 70%.

Farmers Market Analytics System | *Tableau, Python, ETL, SmartSheets*

- Designed, built, and continuously optimized interactive Tableau dashboards covering vendor performance, attendee behavior, sales trends, and benefits utilization, ensuring clarity and usability for vendors, market attendees, and key stakeholders.